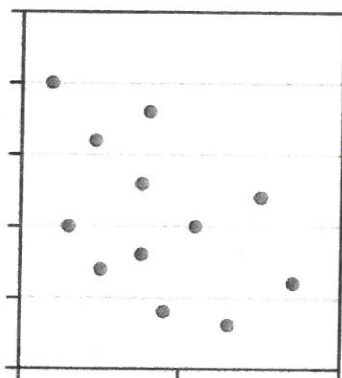


Name:	<i>Mah kindly</i>	
Teacher :		Date: _____
 Baldivis Secondary College	Year 10 General Mini Test 5 Topic – Bivariate Data	
	<u>Full working out MUST be shown to get full marks for each question.</u>	
Total Time:	30 minutes	
Weighting:	5%	
Equipment:	To be provided by the student: Pen, pencil, ruler, scientific calculator, 1 single sided page of A4 notes	

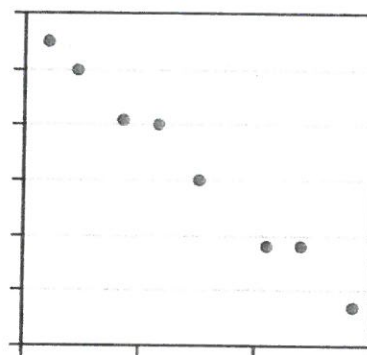
Q1) Identify the shape and direction of the following scatterplots

[6 marks]

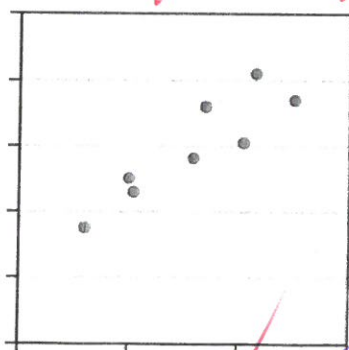
- Use words like positive, negative, linear, non-linear, strong, weak and moderate.



a. *weak Negative*



b. *Strong Negative*



c. *moderate Positive*

Q2) Using your knowledge of Bivariate data

[1 + 2 = 3 marks]

- a. Define the term "Causal Relationship"

when one variable has a direct effect on the other

- b. Provide an example of a causal relationship, and explain how it is causal.

Anything reasonable ✓ *example*
 ✓ *reasoning*

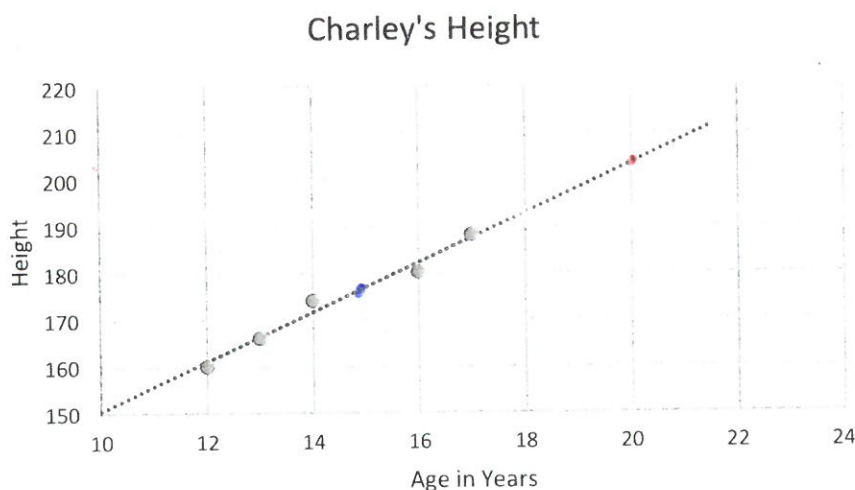
Q3) Identify the independent and dependent variable of the following pairs of bivariate data.

[4 marks]

- a. Number of hours spent fishing and Number of fish caught
 I | D 1/2 each
- b. Hours slept and Achieved test results
 I | D
- c. Ice-cream sales and Daily temperature
 D | I (4)
- d. Price and Average distance run in sneakers
 D | I

Q4) Charley recorded her height from the time she was 12 to 17. She plots a trend line on her data.

[1 + 1 + 2 + 1 = 5 Marks]



a. Using the trend line, make a prediction about how tall she will be at 20 years old?

b. Do you think that your prediction is accurate?
 203 → 207cm
No

c. What do you think would be a reasonable height for Charley at 20 years old? Why?

185 → 190 ish, People grow usually until 18

d. How tall was Charley at 15 years?

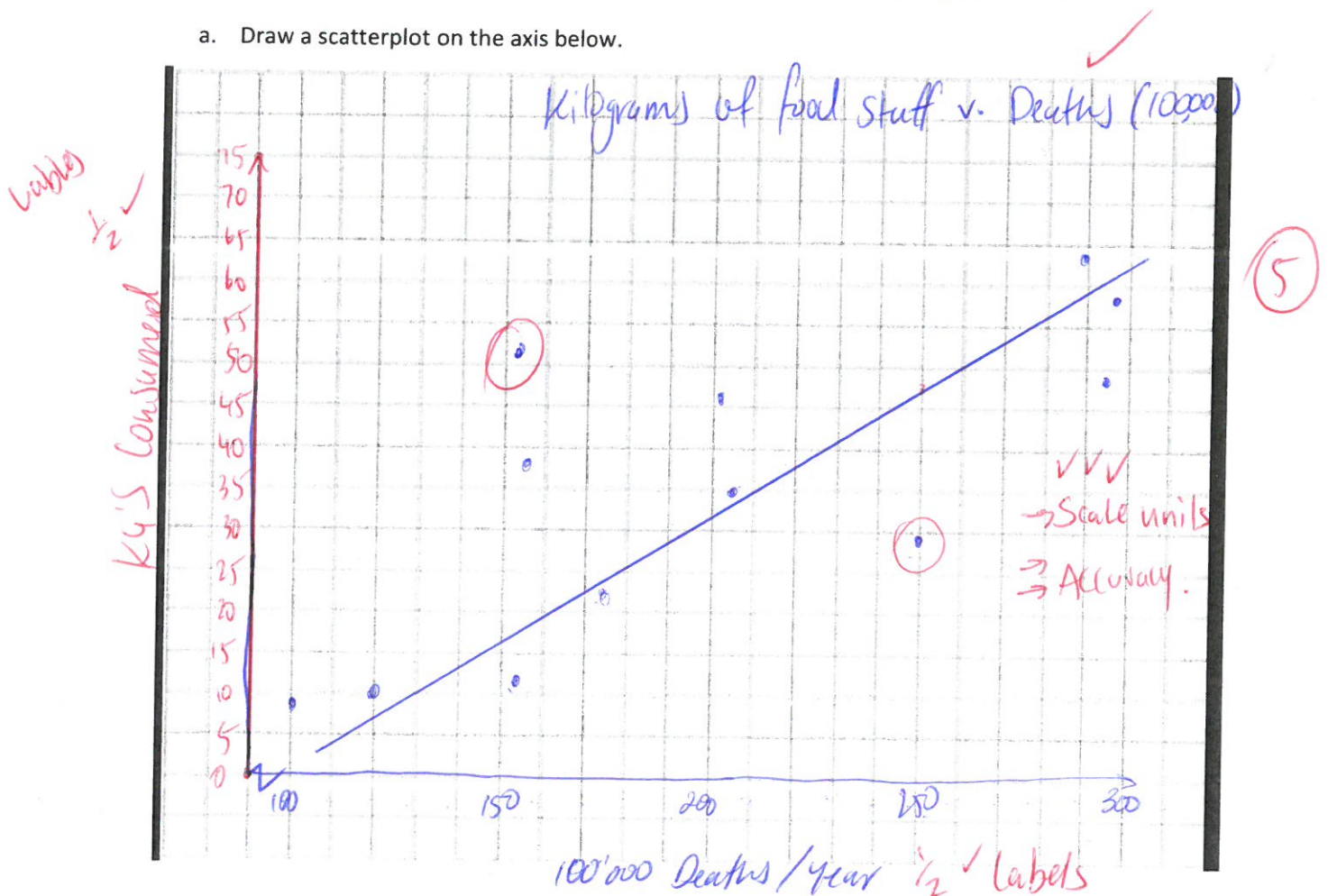
174 → 176cm

Q5) A statistician was asked to investigate whether countries with higher levels of consumption of a particular foodstuff tended to have a higher death rate due to heart disease. The following information was obtained for 12 countries.

[5+2+2+1=10 Marks]

	CONSUMPTION OF FOODSTUFF kg per person per year	DEATHS DUES TO HEART DISEASE Per year per 100 000 of population
Country A	57	299
Country B	13	154
Country C	51	152
Country D	48	295
Country E	35	208
Country F	9	100
Country G	21	175
Country H	10	120
Country I	46	202
Country J	64	290
Country K	38	156
Country L	29	250

a. Draw a scatterplot on the axis below.



b. Fit a trend line to the data

→ Not from zero
→ Straight line

(1)

c. Are there any outliers in this set of data? How could you tell?

250,30 and 152,51, they are removed from LOBF

(2)

d. Krystal says that a country where people eat 50kg's of food would have 250,000 deaths in a year. Does this seem reasonable?

✓ Yes. The trend line interpolation for this is not too far away

(2)

✓ Valid reason

~END OF TEST~